



Modeling a fully electrified passenger Transport System In Ethiopia by 2050



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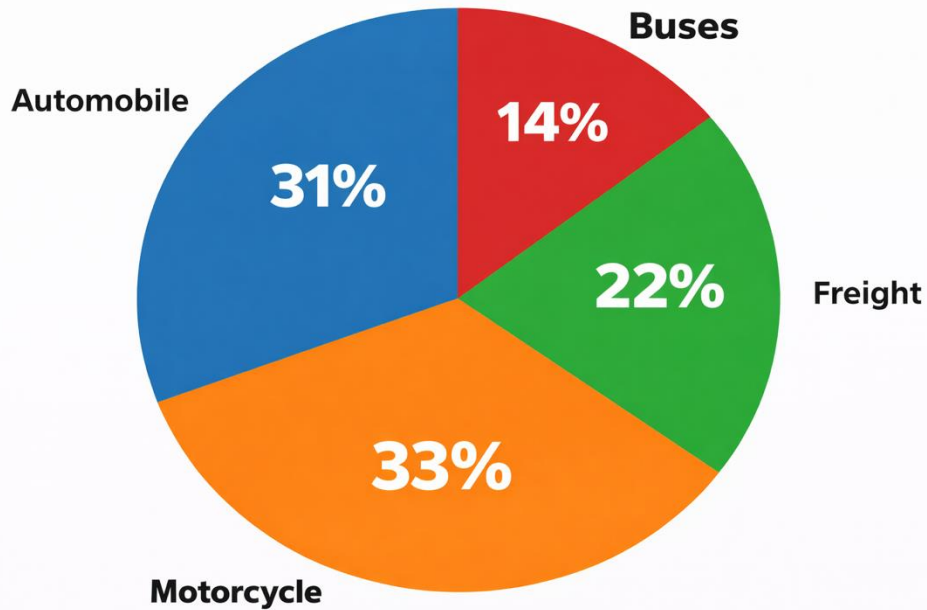
Energy Modelling Platform Ethiopia (EMP-E)

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Outline

- Context
- challenges
- Research Question
- Ethiopia Reference Energy System
- Scenarios
- Results
- Conclusions, Policy insights, & Future Work

Context



The transport sector of Ethiopia covers:

- ✓ 1% of total electricity, 3% of emissions, and of these 20% comes from passenger vehicles (Nnene et al,2025; MoT,2018).
- ✓ Bans some commercial vehicles from JAN, 2024.
- ✓ The government plan shows an increase in EVS from 100,000 to 440,000 by 2030 (Ecofin Agency).

Challenges, and Main Findings

Main challenge(s)

- ✓ High Dependence on Fossil Fuels
- ✓ Rising Energy Demand and Inefficiency
- ✓ Infrastructure inefficiency
- ✓ Environmental Impact and Emissions
- ✓ Infrastructure and Technological Limitations

Main Research Questions

1. How can Ethiopia change the transport system into 100% electrification by 2050?
2. What is the expected electricity demand of Ethiopia's transport sector under a 100% electrification scenario by 2050?

The Main Findings:

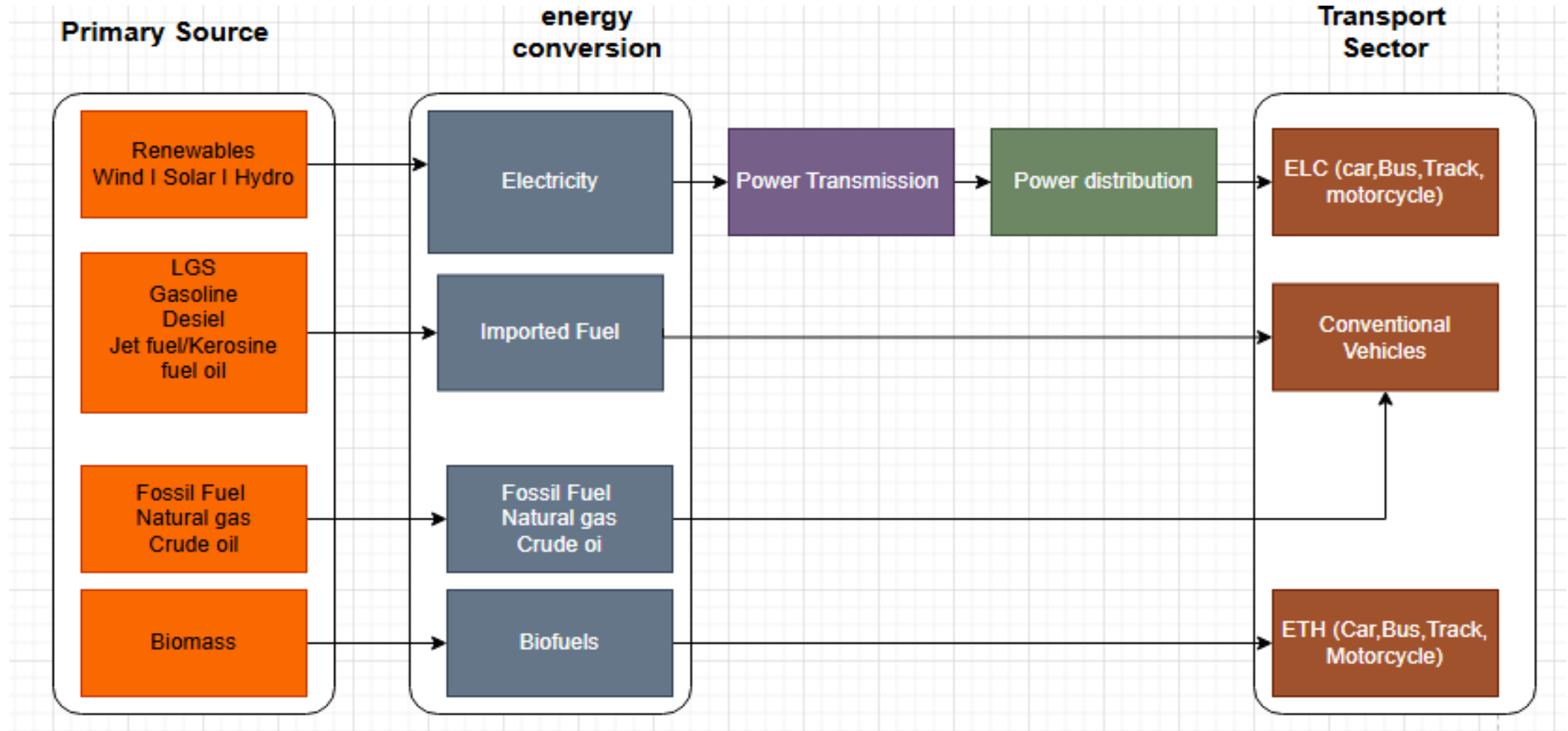
BAU

- High gasoline and diesel demand
- Increased CO₂ emissions
- Highest fuel import costs

100% Electrification

- Increased electricity-based transport
- CO₂ emissions decreased
- Costs mainly associated with electrification

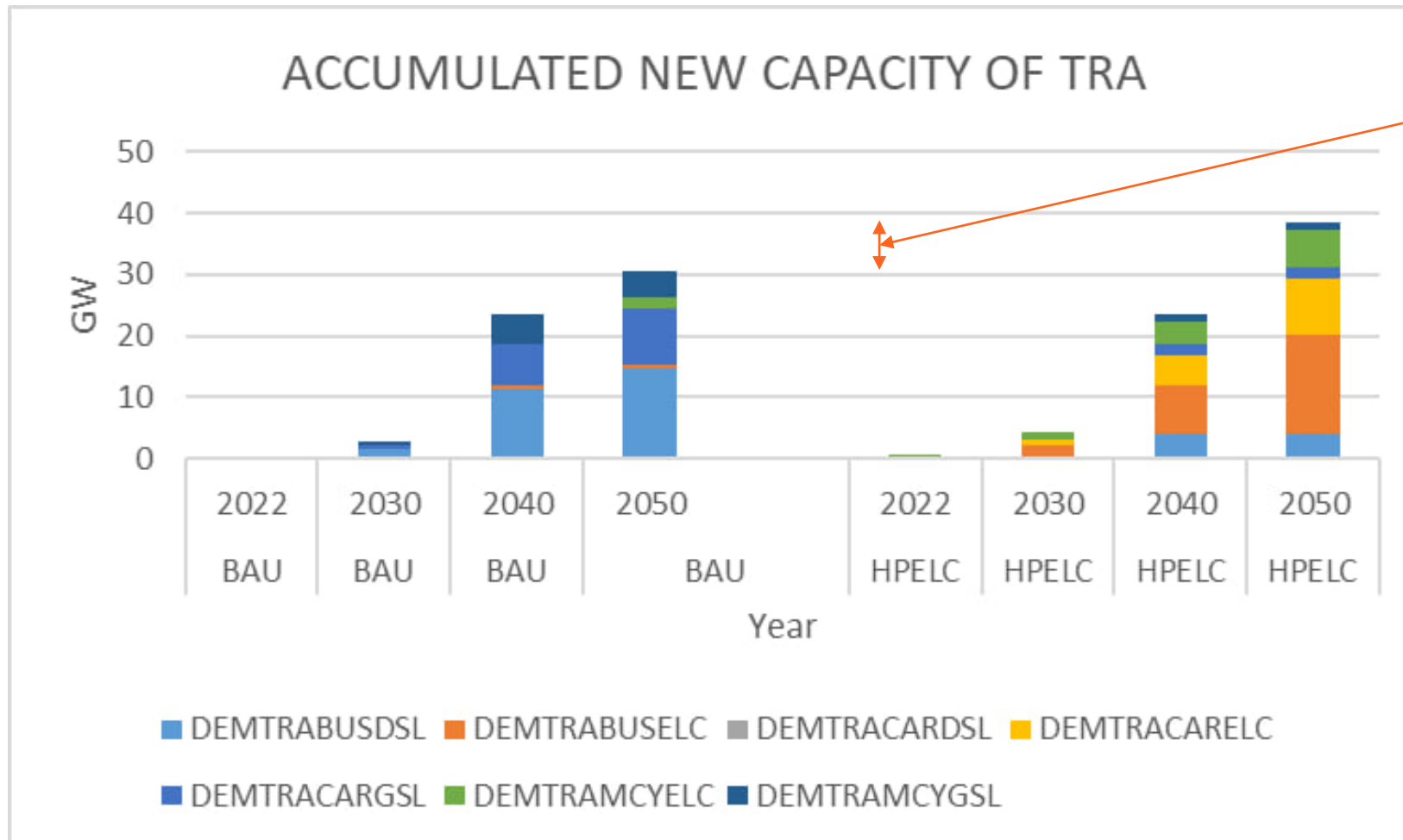
Ethiopia Reference Energy System For Transport Sector



Scenarios

Using the OSeMOSYS Linear Programming Energy System Model, the following scenarios were investigated:

Scenario Label	Scenario Description	Key Assumptions
BAU	Continue current trends without major policy changes.	✓ Currently uses diesel and gasoline; EV adoption is low (7–8 % by 2025) ✓ Increase CO2 emissions
100% Electrification	Electrified the passenger transport system	✓ Changing the passenger transport system into 100% electrification ✓ Existing gasoline and diesel vehicles are gradually becoming obsolete and replaced by EVs. ✓ Investment cost shifts to a renewable source

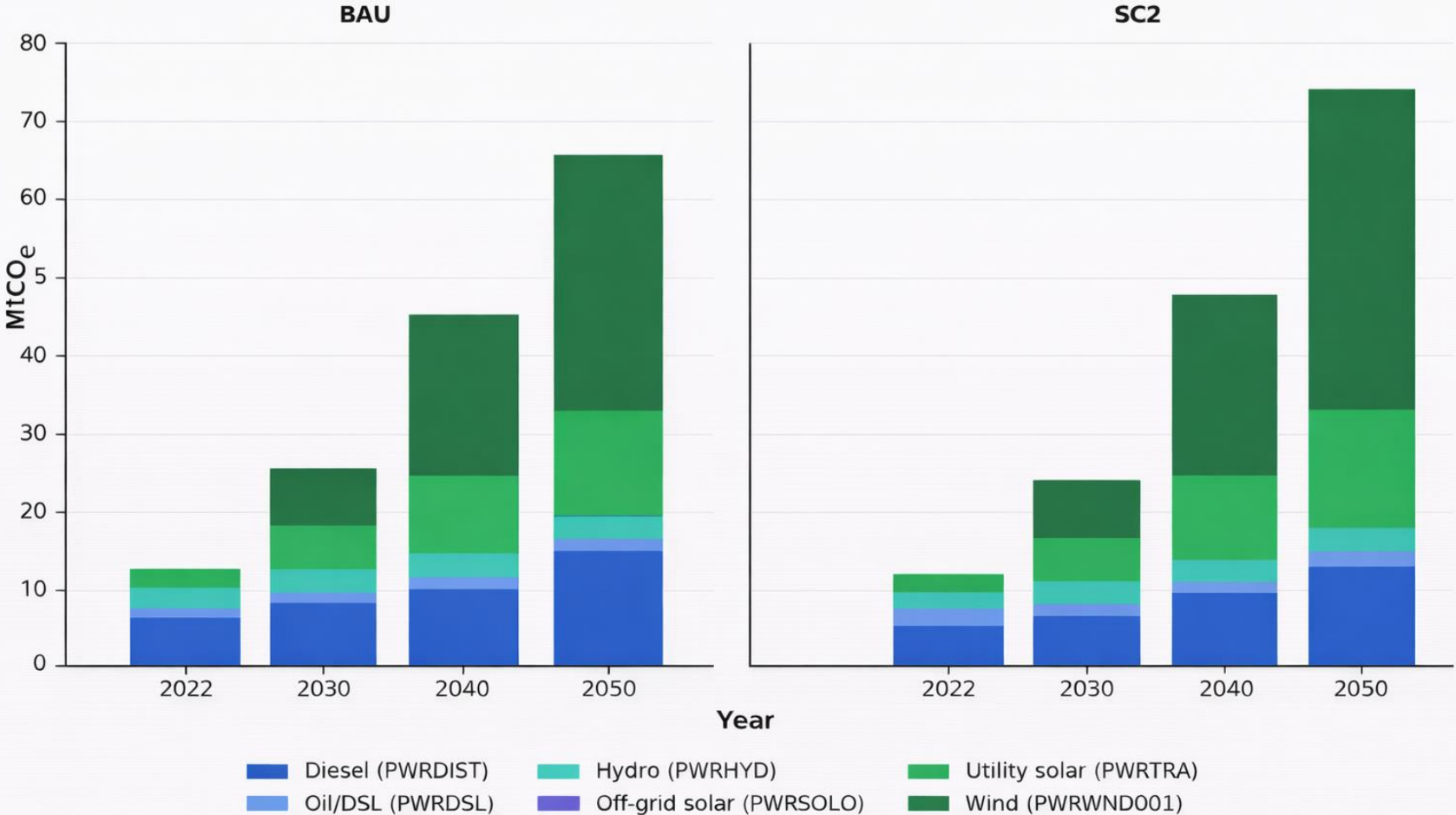


7Gw

☐ In BAU Increase commercial VS

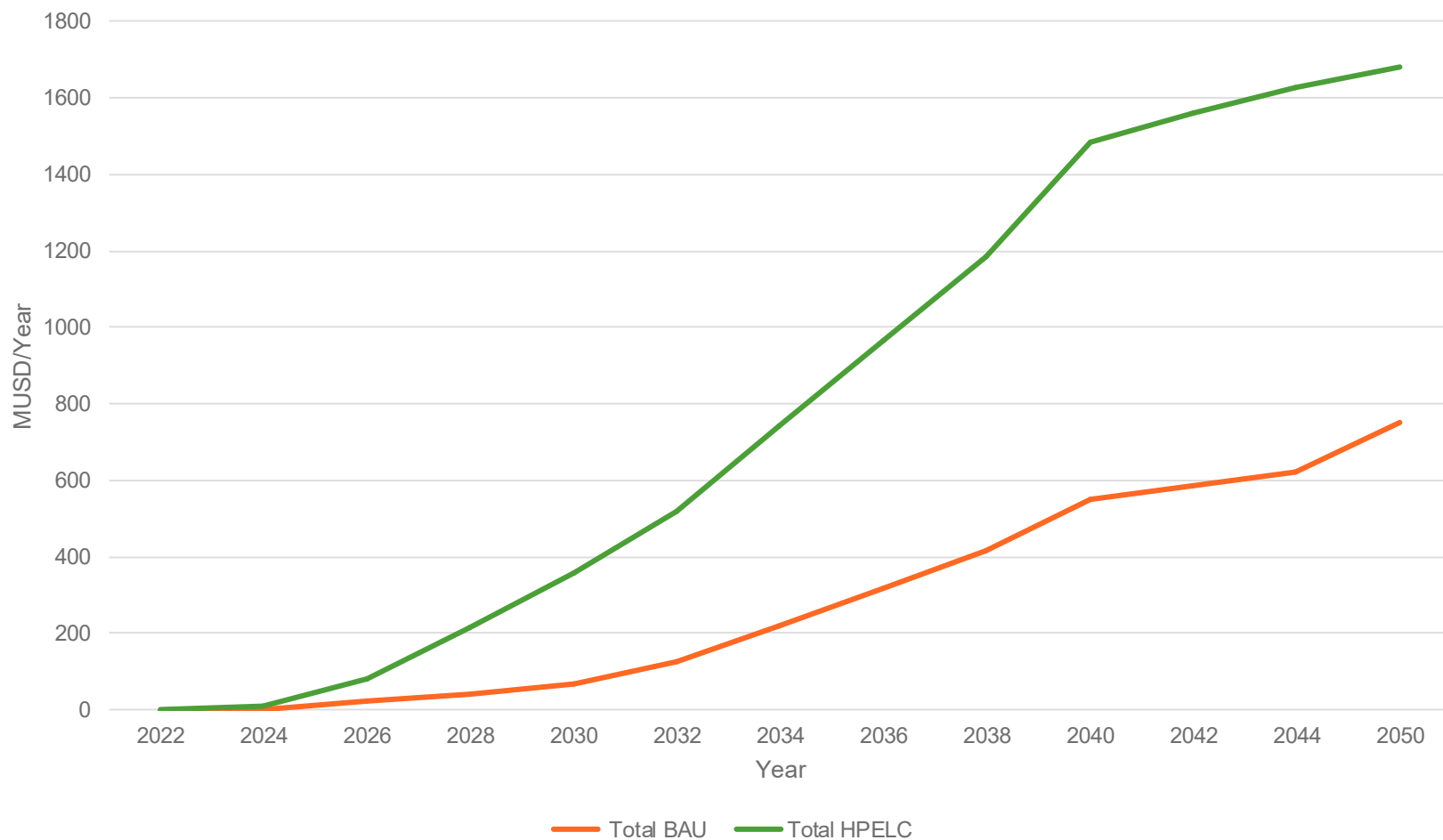
☐ Increase the ELC VS demand

Production by Technology



3 MtCO₂e
averagely increase
renewable energy
in our scenario
than BAU

Annualized Investment cost



➤ Due to a change in commercial VS to EVS and our demand increase, our scenario has higher annualized investment cost.

PRODUCTION BY TECHNOLOGY BY MODE



ANNUAL TECHNOLOGY EMISSION



❖ Total annual technology emission increase on blended diesel and gasoline, and decrease at fully electricity vehicles.

Results

- Annualized Investment Cost increased, especially for electric cars and buses.
- Annual Technology Emission for blended diesel and gasoline is increased in the BAU scenario, but decreased becomes zero for 100% electrification.
- Overall CO2 emission increased with demand for the BAU but decreased for 100% electrification scenario.
- Road transport remains dominant.
- Highest fuel import costs for BAU, but the fuel cost shifts to the electric power investment cost for 100% electrification.

Conclusions

- ✓ The BAU scenario shows that the Ethiopian transport sector largely depends on fossil fuels.
- ✓ This leads to an increase in energy demand and CO2 emissions through time.
- ✓ Due to this reason we need to use additional scenario, so, 100% electrification fills this gap.

Policy Insights

- ✓ ICE ban enables a full electrification pathway by 2050.
- ✓ Charging infrastructure and grid expansion are critical.
- ✓ Renewable electricity ensures low-carbon transport.
- ✓ Affordability and public transport electrification drive success.

Future Work

- ❑ Optimize EV adoption pathways using detailed fleet and infrastructure data.
- ❑ Assess grid integration, peak demand, and storage requirements for full electrification.
- ❑ Include multi-modal transport in future scenarios.
- ❑ Analyze socio-economic impacts and equity of electrification policies.
- ❑ Apply 100% electrification towards the 2050 scenario
- ❑ Invest on low carbon emission technologies
- ❑ Do on large-scale electrification and build infrastructure for this